

ConsMan

Education Consultancy CRM & Student Admission Management Platform

Product Requirements Document (PRD)

Version 1.5 — Development-ready baseline

Prepared: October 2026

Product scope: internal platform for The Blessing Edu; not a multi-agency SaaS product.

Primary objective	Replace fragmented Excel-based lead management with a single source of truth, then manage the complete student journey from lead to enrollment.
Core principle	One person = one master Person record. Stage, status, applications, documents, tasks, communications and history all attach to it.
Primary users	Super Admin, Branch Manager, Counselors, Documentation Staff, Finance Staff, Management; Students through a later portal.
Stack	Next.js + TypeScript + Tailwind; Django + DRF; PostgreSQL; S3-compatible object storage.
MVP	Phase 1: CRM, system-generated QR codes, OTP-verified visitor intake and Excel replacement.

What changed in v1.5

- **Canonical intake boundary:** the IntakeSubmission is stored before OTP delivery; a Person is created, matched or updated only after successful verification. A temporary Person is never created for an unverified visitor. Wording corrected everywhere.
- **Operations:** business calendar, configurable SLA rules, a specified assignment engine with unavailable-owner rules, and a cached next action per Person.
- **Clear boundaries:** Activity vs Communication vs Audit Event defined, plus a unified Person timeline assembled as a read model.
- **Identifiers and hygiene:** reference numbers for Application, Visa Case and Payment; explicit ACTIVE/ARCHIVED rules; Idempotency-Key on all high-risk writes.
- **Reporting foundations:** Source → Campaign → QR hierarchy with a full funnel; configurable lost-reason master.
- **Safety:** EXACT duplicate tier and an explicit never-auto-merge rule; import batch lifecycle with row-level traceability.
- **Phase 2 foundations specified now:** CourseOffering hierarchy, versioned WorkflowTemplates per destination, deadlines and blockers.
- New: FR-41 to FR-43, AC-42 to AC-50, decisions D-21 to D-25.

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1. Executive Summary

- ConsMan is an internal CRM and admission-management system for The Blessing Edu.
- Its immediate purpose is to replace Excel-based lead management that causes duplicate records, missed follow-ups, missing history, inconsistent information and poor visibility.
- It will progressively manage counseling, universities, courses, applications, documents, offers, deposits, CAS/CoE/I-20/LOA, visa processing, pre-departure and enrollment.
- It is not a SaaS product and needs no agency/sub-agent portals or multi-agency management.
- It is person-centric: one Person record persists from first contact (including QR visitor registration) through enrollment and can hold multiple applications.

2. Problem Statement

- Lead and student data is spread across spreadsheets and paper visitor forms.
- Duplicate entries cause multiple staff to contact the same prospective student.
- Follow-up dates are easy to miss and overdue work is hard to identify.
- There is no reliable, centralized interaction history.
- Management cannot see the lead-to-enrollment pipeline.
- Application, document and visa progress is difficult to track consistently.
- Excel cannot enforce ownership, deadlines, verification or audit history.

3. Product Vision & Goals

Build one operational system where every person has a persistent 360° record and every staff action, follow-up, application, document, deadline and outcome is traceable from first lead through enrollment.

- Single source of truth for leads and students; prevent and surface duplicates.
- Explicit ownership and next actions; complete activity and communication history.
- Standardize workflows without forcing every destination into one process.
- Operational visibility for management; fewer missed deadlines.
- A foundation that can later support automation and AI.

4. Success Metrics

Targets are proposed starting points; confirm against the Excel baseline before go-live.

Metric	Definition	Target
Excel retirement	Share of active leads managed only in ConsMan after cutover.	100% within 30 days
Duplicate rate	Duplicate Person records per 100 new leads after cleanup.	< 2%
Overdue follow-ups	Open follow-ups past due ÷ all open follow-ups (weekly).	< 10%, trending down
Time to first contact	Median time from creation (including QR intake) to first logged contact.	< 24 business hours
Intake processing	QR submissions triaged (assigned or merged) within 2 business hours.	> 95%
SLA compliance	SLA timers satisfied before their due time ÷ all timers.	> 90%

Metric	Definition	Target
Intake verification	QR submissions verified by OTP ÷ all QR submissions (verified + unverified).	> 85%
Data completeness	Active leads with phone, source, branch and counselor.	> 98%
Staff adoption	Active staff logging at least one activity per working day.	> 90%

5. Scope & Non-Goals

In scope

- Internal CRM, Person management and student lifecycle; branches, users, roles and permissions.
- System-generated customizable QR codes, OTP-verified visitor registration and staff-assisted intake.
- Lead assignment, duplicate detection, lifecycle, counseling, follow-ups, tasks, activities and notifications.
- University, campus, course, CourseOffering, intake and scholarship data; applications and document requirements.
- Offer and condition tracking; basic deposits/payments (Phase 3) and full finance (Phase 4).
- Enrollment documents (CAS/CoE/I-20/LOA), visa cases, pre-departure and enrollment.
- Dashboards, reports, exports, audit logs; student portal in a later phase.

Explicitly out of scope

- Agency/sub-agent management and agency portals; multi-tenant SaaS; selling ConsMan commercially.
- ERP/accounting beyond consultancy payments and institution-commission tracking.
- Uncontrolled AI decisions on admissions, finance or visa eligibility.

6. Assumptions & Expected Scale

Placeholders for sizing; confirm during Sprint 0.

Item	Planning assumption	Status
Internal users	10–40 staff across 1–3 branches	To confirm
New leads per month	300–1,500 (including QR intake)	To confirm
Person records at go-live	Up to 20,000	To confirm from Excel
Documents per person	10–30 files	Planning
Concurrent users	Up to 30 staff; public intake bursts of up to 20 requests/min at fairs	Derived
OTP messages	About 1.5 OTPs per submission (resends included); monthly provider spend cap set in Settings	To confirm
Time zone / hours	Asia/Kathmandu (NPT); business hours	Decided
Languages	English UI; Unicode (including Devanagari) data entry	Decided (D-05)

7. Decision Log

These decisions close the open questions of v1.2. Each default is adopted so development can proceed; management may change one, and the Impact column says what would be affected. None blocks Sprints 0–2.

ID	Decision (adopted default)	If changed
D-01	Conversion trigger: a counselor (or manager) explicitly clicks 'Convert to student' when the person commits to proceed. Creating the first Application on a LEAD prompts conversion. No manager approval needed to convert.	Conversion action and one permission rule
D-02	Masked discovery: a counselor searching a person owned by someone else sees only: exists, owner name, branch, lead status, created date, and a 'Request access' button. No contact details, notes or documents.	Masked view serializer
D-03	Phase 1 capture channels: manual entry, Excel import, QR/web visitor form and the lead-capture API for the website. Facebook/Instagram lead-form sync is Phase 5.	Phase 5 scope
D-04	No payment data in Phase 1. Registration/counseling fees may be noted as an Activity until the Phase 3 payment record.	Pull payments forward
D-05	English UI; Unicode data entry (Nepali names/addresses supported). Nepali-language UI is a future enhancement.	i18n work
D-06	Intake assignment: new QR submissions go to the branch's unassigned queue and notify Branch Managers. A manager (or an optional round-robin rule) assigns; unassigned beyond 2 business hours is flagged and escalated on the manager dashboard.	Assignment rule only
D-07	Master-list ownership: Super Admin owns sources, campaigns, tags, lost reasons, statuses and branches; Branch Managers can propose additions.	Settings permissions
D-08	Retention (proposed, legal to confirm): retention is a configurable policy per record type. Initial values: raw intake payloads purged 90 days after resolution; LOST/closed leads inactive 36 months flagged for anonymization review; student and visa documents retained while the case is active plus a period set by counsel.	Policy values only
D-09	Student states: Person.stage = STUDENT carries student_state ∈ ACTIVE, ENROLLED, DEFERRED, WITHDRAWN, ARCHIVED. The final lead_status stays CONVERTED for history.	Enum values
D-10	Ownership: one primary owner (counselor) per Person plus optional team members (documentation staff, visa officer) with role and scope.	Team-member model
D-11	Lead temperature (HOT/WARM/COLD) is a separate priority field from lifecycle status. Paper-form 'Enrolled/Closed' map to lifecycle states.	One field
D-12	Identifiers and time: UUID primary keys plus human references: Person BE-YYYY-NNNNNN, Application APP-YYYY-NNNNNN, VisaCase VISA-YYYY-NNNNNN, Payment PAY-YYYY-NNNNNN (year of creation; a counter per type and year). Timestamps stored in UTC and shown in NPT. Records are ACTIVE or ARCHIVED; no hard delete through the UI.	Migrations
D-13	Visa refusal: a re-application after refusal is a new VisaCase (attempt_no + 1) linked to the prior case.	Visa model
D-14	One shared QR: a single permanent QR per branch (and optionally per event) is shown to all visitors. It encodes a short, permanent code on the ConsMan domain, never a per-visitor token. Branch/source/campaign are resolved on the server. Restyling a QR never changes its code, so printed material stays valid.	QR model
D-15	OTP on submit: every public submission must be verified with a one-time code sent to the submitted mobile number by SMS or WhatsApp; the visitor chooses the channel (default SMS, WhatsApp when configured). Staff-assisted entry at the front desk is exempt.	Intake flow
D-16	Store first, verify after: each submission is committed to the database before any OTP is sent and is kept whether or not it is verified. Unverified submissions are untrusted: they never create or change a Person, never match or merge, never notify owners, and appear only in the Unverified queue.	Core intake design
D-17	OTP parameters (tunable in Settings): 6 digits; valid 5 min; 5 wrong attempts then locked; resend cooldown 60 s; 3 sends per challenge; 5 sends per phone and 20 per IP per 24 h. Nepal (+977) numbers allowed by default; other country codes by allow-list, otherwise use staff-assisted entry.	Limits only
D-18	Unverified retention: 30 days, then purged unless a manager verified it by call or discarded it earlier.	Policy value

ID	Decision (adopted default)	If changed
D-19	Messaging providers: an SMS gateway with Nepal coverage and a WhatsApp Business provider with an approved authentication template, behind one internal interface. Select and contract before the end of Sprint 3 (template approval can take days).	Vendor integration
D-20	QR styling limits: free styling within safety rules: logo $\leq$ 20% of QR area with error-correction level H; dark modules on light background at contrast $\geq$ 4.5:1 (inverted or low-contrast blocked); brand colors by default; every design must pass an automatic decode test before download.	Safety rules
D-21	Intake boundary rule: an IntakeSubmission is stored before OTP delivery. A Person is created, matched or updated only after successful OTP (or staff) verification. A temporary Person is never created to hold an unverified visitor.	Core intake design
D-22	Assignment algorithm: QR resolves to a branch $\rightarrow$ submission verified $\rightarrow$ branch assignment rule (round-robin among eligible counselors, or manager queue) $\rightarrow$ first follow-up created $\rightarrow$ SLA timers start. Eligibility and unavailable-owner handling are defined in Section 29.	Assignment engine
D-23	Business calendar and SLAs are configuration: default Sunday-Friday 10:00-17:00 NPT with branch holidays (to confirm). Default SLAs: new lead first contact 24 business hours; HOT lead first contact 4; QR lead assignment 2; access-request response 8.	Settings values
D-24	Workflow templates from Phase 2: destination-specific milestones, required documents, offer conditions, enrollment and visa requirements are versioned configuration, never hard-coded country logic.	Phase 2 design
D-25	Next action is a projection: it is cached on Person from the earliest open FollowUp/Task; it is not a separate source of truth.	Cached fields

8. User Roles & Permission Matrix

Role	Primary responsibilities	Default scope
Super Admin	Configuration, users, branches, workflows, master data, full oversight.	Full
Branch Manager	Branch leads, workload, intake triage, reassignment approval, reports.	Own branch
Counselor	Own leads, counseling, follow-ups, applications.	Own records; masked discovery
Documentation Staff	Document collection, verification, application and visa preparation.	Assigned records
Finance Staff	Payments, deposits, receipts, institution commission.	Finance scope
Management	Dashboards and reports.	Read; configured exceptions
Student	Own status, requested documents, tasks (later phase).	Own profile only

Phase 1 permission matrix (implemented as configuration)

Full = all records · Branch = own branch · Own = assigned to the user · Assigned = user is a team member · Read = read only · — = no access. The matrix must live in configuration/seed data, not scattered through code.

Action	Super Admin	Branch Mgr	Counselor	Docs	Finance	Mgmt
View person details	Full	Branch	Own	Assigned	—	Read
Masked discovery of others' persons	n/a	n/a	Yes	Yes	—	n/a
Create person	Full	Branch	Own branch	—	—	—
Edit person	Full	Branch	Own	Assigned (doc fields)	—	—
Change lead status / temperature	Full	Branch	Own	—	—	—
Convert LEAD → STUDENT	Full	Branch	Own	—	—	—
Request access / reassignment	n/a	n/a	Yes	—	—	—
Approve or perform reassignment	Full	Branch	—	—	—	—
Bulk reassign / bulk update	Full	Branch	—	—	—	—
Create / complete follow-ups, tasks, activities	Full	Branch	Own	Assigned	—	—
Triage intake queue	Full	Branch	If assigned	—	—	—
Create / edit / pause QR codes	Full	Branch	—	—	—	—
View unverified submissions	Full	Branch	—	—	—	—
Verify by call / discard unverified	Full	Branch	—	—	—	—
Review duplicates	Full	Branch	Flag only	Flag only	—	—
Merge duplicates	Full	Branch	—	—	—	—
Import Excel	Full	Branch (sign-off)	—	—	—	—
Export	Full	Branch	Own, no sensitive fields	—	Finance	Reports
Sensitive documents (view/download)	Full	Branch	Own	Assigned	Finance docs	—
Dashboards	Full	Branch	Own	Own queue	Finance	Read
View audit log	Full	Branch	—	—	—	Read
Manage users, roles, branches, master lists, SLAs, calendars, workflows	Full	Propose lists	—	—	—	—

Rules: permission checks are enforced on the server for every endpoint and queryset; hiding a button is never sufficient. MFA is required for Super Admin and Finance Staff (Should for others). Students (Phase 6) access only their own Person, Applications and Documents.

9. Core Data Model

The model separates the **person** from applications. A single **Person** record has a stage (LEAD or STUDENT); conversion changes the stage on the same record, so nothing is copied or re-linked. All tables carry created_at, updated_at, created_by and archived_at (soft delete).

```
PERSON (stage: LEAD → STUDENT)
├ ContactMethod · ConsentRecord · EducationRecord · TestScore · PersonTag · TeamMember
├ Activity · FollowUp · Task · Notification · OwnershipHistory
├ IntakeSubmission (origin)
├ Application(s)
├ Document(s) · Offer · Payment(s) · EnrollmentDocument(s)
├ VisaCase(s) (attempt_no) → Enrollment / PreDeparture
```

Architectural boundary: visitor vs Person

```
Public Visitor → IntakeSubmission (UNVERIFIED) → OTP Verification → Duplicate Detection → Person → CRM Lifecycle
```

The IntakeSubmission and the Person are different records with different trust levels. **A temporary Person is never created to hold an unverified visitor** (D-21). Staff-assisted entry follows the same pipeline with staff verification.

Entity	Key fields and notes	Phase
User / Role / Branch	User(role, branch, is_active, mfa), Role(permission set), Branch(code, name). Branch code is used in QR links.	1
Person	id (UUID), ref (BE-YYYY-NNNNNN), full_name, stage, lead_status, temperature, student_state, lost_reason_id, hold_until, owner_id, branch_id, source_id, dob, address, guardian_name, guardian_contact, preferred_countries[], study_levels[], preferred_course, preferred_intake, preferred_university, previous_visa_refusal (YES/NO/UNKNOWN), highest_education, work_experience, best_contact_method, next_action_type, next_action_due_at, next_action_owner_id (cached from the earliest open FollowUp/Task), last_activity_at, converted_at, record_status (ACTIVE / ARCHIVED), import_batch_id, import_row_number, source_file.	1
ContactMethod	person_id, type (PHONE/EMAIL/WHATSAPP/MESSENGER), raw_value, normalized_value, is_primary, is_shared, shared_with_note, verified_at, verified_via (OTP_SMS / OTP_WHATSAPP / STAFF_CALL / STAFF_IN_PERSON / NONE). Unique index on normalized phone/email for lookup (not unique constraint).	1
TeamMember	person_id, user_id, role_on_case (DOCUMENTATION/VISA/OTHER), active.	1
BusinessCalendar / Holiday	BusinessCalendar(branch_id or global, timezone Asia/Kathmandu, working_days, open_time, close_time); Holiday(date, name, branch_id).	1
SlaRule / SlaTimer	SlaRule(name, trigger, target_business_hours, scope, active); SlaTimer(object, rule_id, started_at, due_at, paused_at, breached_at, satisfied_at). Due times are computed from the BusinessCalendar.	1
AssignmentRule / UserAvailability	AssignmentRule(branch_id, mode ROUND_ROBIN / MANAGER_QUEUE, eligible_roles, max_open_leads, active); UserAvailability(user_id, status ACTIVE / ON_LEAVE / INACTIVE, from_date, to_date).	1
OwnershipHistory	person_id, from_user, to_user, reason, changed_by, changed_at, request_id.	1
SourceMaster	Source → Campaign → Source Detail (three linked master tables with codes). Person stores the leaf id.	1
Tag / PersonTag	Configurable supplementary labels.	1
ConsentRecord	person_id, channel set, given_at, wording_version, method (WEB_FORM/PAPER/STAFF), ip/user-agent for web, withdrawn_at.	1
EducationRecord	person_id, level, institute, degree_stream, grade_or_percent, passed_year.	1

Entity	Key fields and notes	Phase
TestScore	person_id, test (IELTS/PTE/TOEFL/DUOLINGO/SAT/GRE/GMAT/OTHER), score, status (TAKEN/PLANNED/NOT_TAKEN), test_date.	1
IntakeSubmission	id, qr_code_id, payload (JSON), phone_e164, channel_choice, status (UNVERIFIED / VERIFIED / PROCESSED / DISCARDED / PURGED), outcome (CREATED / MATCHED / NEEDS_REVIEW), unverified_reason (NOT_ENTERED / WRONG_OTP / EXPIRED / LOCKED / SEND_FAILED), person_id, resume_token_hash, ip_hash, idempotency_key, verified_at, verified_by_staff_id, purge_after. Committed before any OTP is sent.	1
QRCode	id, code (short, unique, permanent), label, branch_id, campaign_id (campaign → source), content_type (REGISTRATION / WHATSAPP / URL / VCARD), status (ACTIVE / PAUSED / EXPIRED), expires_at, design (JSON: module shape, eye frame, eye ball, colors/gradient, frame and caption, logo ref, error correction), asset_version, scan_count, created_by. Restyling creates a new asset version; the code never changes.	1
OtpChallenge	id, submission_id, phone_e164, channel, code_hash (HMAC with per-challenge salt and server secret; plaintext never stored), expires_at, attempts, max_attempts, send_count, last_sent_at, status (PENDING / VERIFIED / EXPIRED / LOCKED / SUPERSEDED).	1
MessageDelivery	id, challenge_id, provider, channel, to_hash, template, provider_message_id, status (QUEUED / SENT / DELIVERED / FAILED), error, cost, created_at. The OTP value is never logged.	1
DuplicateCandidate	person_a, person_b, confidence (HIGH/MEDIUM/LOW), signals[], status (OPEN/DISMISSED/MERGED), reviewed_by.	1
MergeRecord	survivor_id, merged_id, performed_by, performed_at, field_choices (JSON), moved_counts, reversible_until.	1
AccessRequest	person_id, requested_by, type (ACCESS/REASSIGN), reason, status (PENDING/APPROVED/DENIED), decided_by, decided_at.	1
Activity	person_id, type, channel, direction, outcome, subject, notes, performed_by, performed_at, related (application/task/document), corrects_id. Append-only: corrections create a new row linked by corrects_id; original stays visible in history.	1
FollowUp	person_id, type, due_at, method, owner_id, status, outcome, notes, completed_at. A contact with the person.	1
Task	person_id/application_id (optional), title, priority, due_at, owner_id, status. Internal work (not a contact with the person).	1
Notification	recipient_id, type, title, message, related_type/id, read_at.	1
ImportBatch / ImportRow	batch(file, mapping, status STAGING / VALIDATED / APPROVED / COMMITTED / ROLLED_BACK, counts, created_by, approved_by); row(import_batch_id, row_no, source_file, legacy_id, raw, normalized, result, person_id). Every imported Person stores import_batch_id, import_row_number and source_file.	1
AuditEvent	actor, action, object_type/id, old/new (JSON), at, ip, request_id. Append-only.	1
LostReason / Status lists	Controlled master data owned by Super Admin.	1
Document / DocumentVersion	type, context (person/application/visa), status, expiry, storage_key, version, uploaded_by, verified_by, checksum.	2
University / Campus / Course / CourseOffering / Intake	University → Campus; Course belongs to a University; Intake = term/date window. CourseOffering = Course + Campus + Intake (unique on that triple): the selectable admission opportunity, carrying requirements and fees. Applications reference a CourseOffering only.	2
Application	ref (APP-YYYY-NNNNNN), person_id, course_offering_id, workflow_template_id and version, state, stage history, owner. Independent per application.	2
WorkflowTemplate	destination, version, milestones[], required_documents[], offer_conditions[], enrollment_requirements[], visa_requirements[]. Versioned configuration; an application keeps the version it started with (D-24).	2

Entity	Key fields and notes	Phase
Deadline	entity_type/id, type (APPLICATION / DOCUMENT / OFFER_EXPIRY / DEPOSIT / ENROLLMENT_DOC / VISA_APPOINTMENT / VISA_SUBMISSION / COURSE_START / PRE_DEPARTURE / FLIGHT), due_at, source, status.	2
Blocker	entity_type/id (Person, Application, VisaCase), type, description, assigned_to, created_at, resolved_at.	2
Offer	application_id, type, conditions, expiry, status.	2-3
Payment	ref (PAY-YYYY-NNNNNN), person_id, application_id (optional), amount, currency, date, method, reference, status, proof document. Basic in Phase 3, full in Phase 4.	3
EnrollmentDocument	application_id, kind (CAS/COE/I20/LOA/other), number, issued_at, expires_at, document_id.	3
VisaCase	ref (VISA-YYYY-NNNNNN), application_id, attempt_no, previous_case_id, state, events[], appointment, decision.	3
Enrollment / PreDeparture	Final enrollment outcome, checklist, travel and arrival details.	3

10. Lifecycle & State Machines

Person stage and student state

stage: LEAD → STUDENT

student_state (when STUDENT): ACTIVE → ENROLLED | DEFERRED · WITHDRAWN · ARCHIVED

Stage is a high-level category and never replaces application status. Reverting STUDENT → LEAD requires Branch Manager approval and an audit entry.

Lead status (while stage = LEAD)

NEW → CONTACTED → COUNSELING → INTERESTED → DOCUMENT_COLLECTION → APPLICATION_READY → CONVERTED

Any active status → LOST | ON_HOLD

- Forward moves normally go one step at a time; skipping requires a note. Backward moves require a reason and are audited.
- LOST requires a controlled lost reason. A LOST lead can be reopened by its owner or a manager; this logs an activity and re-runs duplicate detection.
- ON_HOLD requires a resume date and automatically creates a follow-up for that date.
- CONVERTED is set by the conversion action (D-01); it changes stage to STUDENT on the same record. After conversion the lead status is frozen and the person leaves the lead pipeline.
- Leads with no activity for a configurable period (default 30 days) are flagged stale; they are never auto-closed.
- **Temperature** (HOT / WARM / COLD) is a separate, freely editable priority field and does not affect lifecycle.

Lost reasons (configurable master)

Cost / financial constraints · Changed destination · Changed university · Visa refusal concern · Academic eligibility / profile mismatch · Family decision · Not responding / not reachable · Not interested · Chose another consultancy · Applied directly · Delayed study plan · Personal reason · Duplicate / invalid record · Other (note required). Lost reasons are a **configurable master** (active flag, sort order, optional category) managed by Super Admin; a reason is mandatory on every LEAD → LOST transition, and reports group by reason and category.

Application state (Phase 2-3)

DRAFT → DOCUMENT_COLLECTION → READY → SUBMITTED → UNDER_ASSESSMENT → OFFER_RECEIVED → OFFER_ACCEPTED → ENROLLED

Terminal/side states: REJECTED · OFFER_DECLINED · OFFER_EXPIRED · WITHDRAWN · DEFERRED · CANCELLED

- This chain covers the **admission** only. Deposit, enrollment-document and visa progress are **derived milestones** from Payment, EnrollmentDocument and VisaCase and are shown on Application 360; they are not application states, so there is a single source of truth.
- ENROLLED is set when the Enrollment record is confirmed. DEFERRED records the deferral and may re-point the application to a new CourseOffering with history preserved.
- The chain is the default template; destination-specific templates (UK, Australia, USA, etc.) may add or reorder steps through configuration.
- Activation: Phase 2 enables DRAFT through OFFER_RECEIVED and the terminal states; Phase 3 enables OFFER_ACCEPTED and ENROLLED.

Document state

REQUESTED → UPLOADED → UNDER_REVIEW → VERIFIED

UNDER_REVIEW → REJECTED → RESUBMISSION_REQUIRED · VERIFIED → EXPIRED · any → WAIVED / NOT_APPLICABLE

Visa state (Phase 3)

PREPARING → READY → SUBMITTED → BIOMETRICS_PENDING → INTERVIEW (if applicable) → ADMIN_PROCESSING → APPROVED | REFUSED | WITHDRAWN

Additional-information requests are visa events/tasks and do not overwrite case history. After REFUSED, a new VisaCase is created with attempt_no + 1 (D-13).

11. Phase Roadmap

Phase	Focus	Primary outcome
1	CRM, QR Intake & Excel Replacement	System-generated QR codes, OTP-verified visitor intake, Person records, duplicates, follow-ups, history, import and cutover.
2	Admissions Management	Universities, courses, CourseOfferings, applications and structured documents.
3	Offer, Visa & Enrollment	Offers, basic deposits, enrollment documents, visa and pre-departure.
4	Finance & Institution Commission	Full payments, receipts, expenses and institution commission.
5	Communication & Automation	Email/WhatsApp/SMS, templates, ad-form sync, Redis/Celery, automation.
6	Student Experience	Student portal and self-service.
7	AI & Intelligence	Summaries, matching assistance, risk flags, productivity.

12. Phase 1 — CRM, QR Intake & Excel Replacement

Objective: replace Excel (and paper forms) as the operational source of truth for leads and student relationships. M = Must (MVP), S = Should, C = Could.

ID	Requirement	Pri.
FR-01	Create Person/lead manually through a form.	M
FR-02	Generate unique immutable IDs: UUID plus human reference BE-YYYY-NNNNNN for Person (APP-/VISA-/PAY- references for later records, D-12).	M
FR-03	Master profile per Section 22: contacts, preferences, education, tests, guardian, address, visa-refusal flag, source, branch, owner.	M
FR-04	Every active lead has exactly one owner and branch; ownership history retained.	M
FR-05	Duplicate detection with normalized contacts, shared-number handling and confidence levels (Section 23).	M
FR-06	Authorized side-by-side merge preserving all history, with merge record.	M
FR-07	Follow-ups with type, date/time, method, owner, outcome and notes.	M
FR-08	In-app due/overdue follow-up visibility; leads without next action flagged.	M
FR-09	Unified Person timeline on Person 360 (Section 25): activities, follow-ups/tasks, intake and verification milestones, assignment and status changes.	M
FR-10	Manual communication logging (call, WhatsApp, SMS, email, Messenger, meeting).	M
FR-11	Click-to-chat (wa.me) and tel: links that prompt an activity log.	S
FR-12	Public visitor registration (shared QR/web form) feeding the intake pipeline, per Sections 13–15.	M
FR-13	Tasks with owner, priority, status and due date.	M
FR-14	Search by name, phone (any format), email, Person ID and indexed fields.	M
FR-15	Filters: country, counselor, branch, status, temperature, source/campaign, tag, date, follow-up state.	M
FR-16	Excel import with mapping, validation, duplicate review, preview, idempotency, batch lifecycle (STAGING → VALIDATED → APPROVED → COMMITTED → ROLLED_BACK) and row-level traceability.	M
FR-17	Authorized export with audit record and sensitive-field exclusion.	M
FR-18	Lifecycle rules: transitions, lost reasons, hold dates, conversion action.	M
FR-19	Bulk reassign/status/tag updates with audit entries.	S
FR-20	In-app notification center (assignments, intake, overdue work, duplicate reviews, access requests).	M
FR-21	Visibility rules: masked discovery, access/reassignment requests and approvals (Section 29).	M
FR-22	Staff-assisted intake: the same form behind login for front-desk use.	S
FR-23	Lead temperature field (HOT/WARM/COLD) with filter and dashboard counts.	M
FR-24	Structured education records and test scores.	S
FR-25	Role/permission matrix enforced server-side per Section 8.	M
FR-26	Append-only audit trail for critical actions.	M
FR-27	Consent records captured on intake, import and manual entry.	M
FR-28	Source → Campaign → Source Detail masters and configurable tags.	M
FR-29	Counselor, manager and intake dashboards.	M
FR-30	Stale-lead flagging after configurable inactivity.	S
FR-31	QR Studio: create, preview, test, download and manage QR codes inside ConsMan, per branch/source/campaign/event.	M

ID	Requirement	Pri.
FR-32	QR customization: center logo, module/eye/edge styles, colors with brand presets, frame and caption, saved style templates; PNG, SVG and print-ready PDF.	M
FR-33	Multiple QR content types: registration form (default), WhatsApp chat, custom URL, contact card.	S
FR-34	Permanent shared QR: stable short code per branch/event; restyling never changes the code; pause, expire and archive; friendly inactive page.	M
FR-35	OTP verification on submit via SMS or WhatsApp; a Person is created, matched or updated only after successful verification.	M
FR-36	Store-first durability: every public submission is committed before any OTP is sent and kept whether or not verified.	M
FR-37	Unverified Submissions queue with verify-by-call, discard, edit-with-audit and automatic purge.	M
FR-38	OTP abuse and cost protection: Turnstile, per-phone/IP/QR limits, country allow-list, spend cap and alerts, hashed codes, delivery logs.	M
FR-39	QR analytics: scans, submissions, verified submissions and Persons created per QR.	S
FR-40	Messaging provider abstraction (SMS and WhatsApp adapters) with delivery status handling.	M
FR-41	Next action: cached projection of the earliest open follow-up/task shown on lists and Person 360; leads without one are flagged.	M
FR-42	Business calendar and configurable SLA rules driving due times, breach flags and escalations.	M
FR-43	Assignment engine: branch rules, round-robin or manager queue, eligibility checks and automatic handling of unavailable owners.	M

13. Visitor Intake (Shared QR Registration)

Replaces the paper Visitor Registration Form (page 1). Page 2, the Counseling Record, is office-only and is captured inside ConsMan as an Activity (counseling summary), document requests, a FollowUp (next step and date), lead status and temperature.

Canonical rule and boundary

The IntakeSubmission is stored before OTP delivery. A Person is created, matched or updated only after successful OTP verification (or staff verification). These are two different records with different trust levels, and a temporary Person is never created to hold an unverified visitor (D-21).

Public Visitor → IntakeSubmission (UNVERIFIED) → OTP Verification → Duplicate Detection → Person → CRM Lifecycle

Flow

- **1. Scan.** Every visitor scans the **same shared QR** (one per branch or event, D-14) and opens the public form at `register.theblessingedu.com/r/{code}`. Branch, source and campaign come from the QR record on the server, never from editable URL parameters.
- **2. Submit.** The visitor fills in the details and taps Submit. The server validates the data, **commits an IntakeSubmission with status UNVERIFIED**, and only then asks the messaging provider to send a one-time code to the submitted mobile number by SMS or WhatsApp (visitor's choice).
- **3. Verify.** The visitor enters the code on the next screen (Section 15). A correct code marks the submission VERIFIED and the number as verified.
- **4. Process.** Only verified submissions enter the pipeline. **No match:** create Person (stage LEAD, status NEW) with source, branch, ContactMethods, education/test rows, ConsentRecord, an Activity 'Registered via visitor form (verified by OTP)' and a first-contact FollowUp due within 24 business hours; assign per the assignment engine (D-22, Section 29). **High-confidence match:** no second Person; attach the submission as an Activity on the existing record, notify its owner and create a Duplicate

Review item. **Medium match:** NEEDS_REVIEW in the intake queue.

- **5. Not verified.** If the visitor abandons the page, enters wrong or expired codes, is locked out, or the provider fails, the submission stays UNVERIFIED and safely stored. It never creates a Person; authorized staff handle it in the Unverified queue (Section 15).
- The visitor always sees neutral messages, so the form cannot be used to find out whether someone already exists in ConsMan.

Public and staff endpoints

Item	Specification
Public routes	GET /r/{code} (form page). POST /api/public/intake/submissions (validate, store, send OTP; returns submission id and resume token). POST .../{id}/otp/verify · POST .../{id}/otp/resend · PATCH .../{id}/phone (correct the number, limited).
Staff routes	GET /api/v1/intake/submissions?status=... · POST .../{id}/resolve · POST .../{id}/verify-by-call · POST .../{id}/discard.
Abuse protection	Cloudflare Turnstile (or equivalent) before the first OTP send; per-IP, per-phone and per-QR rate limits; honeypot field; payload cap (≤ 32 KB); strict field whitelist; CORS limited to the form origin; no cookies or sessions on public routes.
Idempotency	Client-generated Idempotency-Key header; repeated submits return the original submission instead of creating another.
Validation	Required: full name, mobile (valid E.164 for an allowed country), consent checkbox. Everything else optional. Text stored as plain text and escaped on output.
Consent	Checkbox wording is versioned; ConsentRecord stores version, timestamp and request metadata and becomes valid only when the number is verified.
Data handling	IP stored hashed; no documents uploaded through the public form; raw payload of verified submissions purged per D-08, unverified per D-18.
Failure behavior	If the network fails the form keeps the visitor's answers on screen and retries; a completed form must never be lost.

Field mapping (paper form → ConsMan)

Paper form field	ConsMan target
Full name; mobile; WhatsApp/alternate; email	Person.full_name; ContactMethod (PHONE, WHATSAPP, EMAIL)
Address; guardian name and contact	Person.address; guardian_name, guardian_contact
Highest education; institute; degree/stream; GPA/%; passed year	EducationRecord (and Person.highest_education)
Work experience	Person.work_experience (text)
IELTS, PTE, TOEFL, Duolingo, others; test status; SAT/GRE/GMAT	TestScore rows
Interested country; study level; course; intake; university	Person.preferred_countries[], study_levels[], preferred_course, preferred_intake, preferred_university
Previous visa refusal	Person.previous_visa_refusal (flagged for counselor attention)
How did you hear about us	Source Detail; the QR's campaign and source take precedence
Best way to contact	Person.best_contact_method
Signature / consent sentence	ConsentRecord (checkbox, timestamp, wording version); valid only after OTP verification
Page 2 (office): summary, documents, next step, lead status, remarks	Activity (meeting), document requests, FollowUp, lead_status/temperature, internal-note Activity

Staff-assisted mode (FR-22): the same form and pipeline behind login for the front desk, additionally showing the Counseling Record fields. OTP is not required because staff verify the visitor in person; the contact is stored with `verified_via = STAFF_IN_PERSON`.

14. QR Management (QR Studio)

QR codes are created, styled, tested, downloaded and managed inside ConsMan (Settings → QR Codes). No external QR generator is needed, and every QR is tied to a branch, source and campaign so attribution stays accurate.

One shared, permanent QR

- One QR per branch (and optionally per event or campaign) is displayed on posters, desk stands and screens and **presented to every visiting student**. It carries no personal or per-visitor data.
- The QR encodes `https://register.theblessingedu.com/r/{code}`: a short, permanent code on the organization's own domain. The form opens directly; there is no third-party shortener and no `claud.ai` or vendor address.
- **Design and destination are separate.** Restyling a QR creates a new asset version but never changes its code, so previously printed material keeps working. If the form moves, only the server mapping changes.
- **Status:** ACTIVE, PAUSED or EXPIRED (optional expiry for events). An inactive code shows a friendly 'This registration link is not active, please see our front desk' page and accepts no submissions.
- Scans are counted (a counter only; no visitor data) for the analytics in Section 26.

Attribution hierarchy

```
Source (Walk-in · Facebook · College visit · Education fair)
├─ Campaign (e.g. BBA UK September 2027)
└─ QR Code (e.g. Education Fair – Kathmandu – Booth 12)
```

- Every QR belongs to exactly one Campaign, and every Campaign to one Source; the branch is set on the QR as well. A default campaign 'Walk-in — {branch}' is created per branch for desk QRs.
- Submissions and Persons store the campaign (source derived) and the QR id, so reports drill from Source to Campaign to QR.
- Funnel per QR: scans → submissions → verified → Persons → applications → enrollments (the last two from Phases 2–3).

QR types

Type	Payload	Priority
Visitor registration (default)	Public form URL for the QR's code.	Must
WhatsApp chat	wa.me link to the office number with an optional prefilled message.	Should
Website / custom URL	Any validated https URL (e.g. a landing page or fair page).	Could
Contact card	vCard with office name, phone, email and address.	Could

Customization options

Element	Options
Center logo	ConsMan mark by default, or an uploaded PNG/SVG logo; sized automatically ($\leq 20\%$ of QR area) with clear padding; can be turned off. Error-correction level H is applied automatically when a logo is present.
Module (dot) style	At least five: square, rounded, dots, extra-rounded, classy/diamond.
Eye frame (finder outer)	At least four: square, rounded, circle, leaf.
Eye ball (finder inner)	At least four: square, rounded, circle, diamond.
Colors	Brand presets (Deep Navy #1E3A8A, Emerald #059669, Slate #0F172A) plus custom colors; optional gradient on modules; background white or transparent.
Frame / edge design	None, rounded border, caption banner (e.g. 'Scan to register', above or below), or a poster card with logo, branch name and caption; border thickness and corner radius adjustable.
Output	PNG at selectable sizes, SVG (vector, for print), and print-ready PDF (A4 poster, A5 flyer, table-tent). Quiet zone of at least 4 modules is enforced.
Saved styles	Reusable style templates (for example 'ConsMan Standard') so every branch's QR matches the brand guidelines; live preview while editing.

Scan-safety rules (D-20)

- Before a download is enabled the server renders the final image, decodes it and compares the payload; a design that does not decode is blocked with the reason shown.
- Blocked automatically: contrast below 4.5:1 between dark and light, inverted (light-on-dark) codes, logo above 20% of the area, quiet zone below 4 modules.
- Warn when the intended printed size is below 25 mm (brand print minimum). A 'Test scan' panel shows what a phone will open.

Management

- QR list with label, branch, type, status, scans, submissions and verification rate; clone, pause, expire, archive; re-download any past asset version.
- Creating, editing, pausing and archiving a QR are permission-controlled (Section 8) and audit logged.

15. Visitor Verification (OTP)

Principles

- **Store first, verify after (D-16).** The submission is committed before any provider call, so provider outages, closed tabs and wrong codes never lose a visitor's data.
- **Untrusted until verified.** Anyone can type any number into a shared QR form, so unverified data must not touch real records.
- **Verification proves control of the number** (not identity) and sets `ContactMethod.verified_at`.
- The OTP is a transactional message sent only when requested, within the limits below.

Visitor experience

- After Submit: 'We sent a 6-digit code to +977 98XX XXX XXX by SMS/WhatsApp.' Actions: enter code (one-time-code autofill supported), Resend (after 60 s), Switch channel, Change number.
- Wrong code shows attempts left; expired or locked states say 'Your details are saved; our team can contact you to confirm.' A send failure says the same and offers the other channel.

- Success: neutral thank-you message with the branch's next steps. The visitor can resume after a refresh with a resume token (valid 24 h).

OTP specification

Item	Specification
Code	6 digits from a cryptographically secure generator; single use; valid 5 minutes; a new challenge supersedes the old one.
Storage	Only an HMAC-SHA256 hash (per-challenge salt plus server secret); constant-time comparison; plaintext never stored or logged.
Attempts	5 wrong attempts per challenge, then LOCKED; counters updated atomically in PostgreSQL.
Resend	60 s cooldown; 3 sends per challenge; switching SMS ↔ WhatsApp counts as a send.
Limits	5 sends per phone and 20 per IP per 24 h; per-QR and global daily caps; provider monthly spend cap with alert to Super Admin and automatic throttling.
Pre-checks	Turnstile token before the first send; honeypot; valid E.164 and allowed country code.
Wrong number	Up to 2 corrections before verification; each creates a new challenge; original values stay in audit; the previous number is not contacted again.
Message	Short transactional text: organization name, code, validity, and 'ignore if you did not request this'. WhatsApp uses an approved authentication template; no marketing content.
Providers	Behind one MessagingProvider interface (SMS gateway with Nepal coverage, WhatsApp Business provider, per D-19); delivery status stored in MessageDelivery.
Failure handling	Provider error or timeout leaves the submission UNVERIFIED (SEND_FAILED) with the visitor told details are saved; managers are alerted when the failure rate is high.
Phase boundary	Phase 1 includes OTP delivery only; templates, two-way messaging and automation remain Phase 5.

Verified vs unverified handling

Aspect	Verified	Unverified
Person record	Created or matched via the Section 13 pipeline.	Never created, changed or linked.
Duplicate detection	Runs normally.	Excluded; never used to reveal that a Person exists.
Consent	ConsentRecord valid.	Recorded on the submission only; not valid for marketing or follow-up beyond one staff verification call.
Notifications	Owner/branch notified.	Managers see counts only; personal data appears only inside the queue.
Who can see it	Normal CRM permissions.	Super Admin and the branch's Branch Manager only (Section 8).
Staff actions	Normal workflow.	Verify by call (reason required, audited; sets verified_via = STAFF_CALL and continues as verified), edit fields (audited), or discard (reason).
Retention	Person lifecycle.	Purged after 30 days (D-18); only counts are kept for metrics.

Isolation of unverified data prevents hijacking: someone who types another person's number into the shared form cannot alter, merge into or learn anything about that person's record.

16. Phase 2 — Admissions Management

- **Hierarchy:** University → Campus; Course; Intake; **CourseOffering = Course + Campus + Intake** (unique), the selectable admission opportunity. Applications reference a CourseOffering only, so course and intake data is never duplicated.

- **WorkflowTemplate per destination:** milestones, required documents, offer conditions, enrollment requirements and visa requirements as versioned configuration, not code (D-24).
- **Deadlines as first-class data** (application, document, offer expiry, deposit, CAS/CoE/I-20/LOA, visa appointment and submission, course start, pre-departure, flight) feeding dashboards and an upcoming-deadlines queue.
- **Blockers** (type, description, entity, assignee, created/resolved) on Person, Application and Visa Case, shown on dashboards.
- University, campus, course and CourseOffering data with level, duration, fee, requirements, English requirements and availability.
- Intake management with deadlines; scholarship information and eligibility; student preference profile; rule-based course matching.
- Multiple applications per Person; configurable stages by destination; application-specific document checklist.
- Document upload, verification, rejection, replacement and versioning; application timeline and task generation.
- Offer receipt with type, conditions, expiry and acceptance status.

17. Phase 3 — Offer, Visa & Enrollment

- Offer condition tracker; deposit requirement and **basic payment record** (amount, currency, date, method, reference, status, proof file).
- Generic enrollment-document model (CAS, CoE, I-20, LOA and others).
- Country-specific visa workflows, checklists, appointments, submission, decision, refusal and additional-information handling.
- Financial-document metadata and verification; pre-departure checklist; accommodation and airport pickup; travel and arrival; final enrollment.

18. Phase 4 — Finance & Institution Commission

- Student invoices/charges where applicable; full payment lifecycle building on Phase 3 records; receipts and references.
- Payment status: pending, received, verified, refunded/cancelled; consultancy operating expenses.
- Institution commission records with configurable calculation rules and received/pending tracking; finance dashboards and exports.

Institution commission means commission received by The Blessing Edu from universities/institutions. Agency/sub-agent commission is out of scope.

19. Phase 5 — Communication & Automation

- Templates; email integration; full WhatsApp and SMS messaging beyond OTP (the provider interface is built in Phase 1); Facebook/Instagram lead-form sync.
- Two-way communication history; Redis + Celery for outbound messaging, webhooks and scheduled automation.
- Automated reminders and workflow notifications for staff and students; per-channel consent enforcement.

20. Phase 6 — Student Experience

- Secure student login; 360° status view; application progress; document upload and requested actions.
- Offer, condition and visa visibility; pre-departure checklist; secure messaging; mobile-responsive experience.

21. Phase 7 — AI & Intelligence

- AI-assisted summaries, course-matching assistance, document completeness checks, follow-up prioritization, timeline summaries and drafting of routine communications.
- Management analytics and anomaly detection. AI must surface evidence and remain reviewable; final admissions, financial and visa decisions stay human-controlled.

22. Person Profile & CRM Fields

Field	Required?	Notes
Person ID / reference	Yes	System-generated, immutable.
Full name	Yes	Unicode allowed; primary display name.
Phone(s)	Yes (≥ 1)	Raw + E.164; primary flag; shared flag with note.
Email(s)	Preferred	Lowercase normalized; multiple allowed.
Date of birth	Optional	Secondary duplicate signal.
Address; guardian name and contact	Optional	Guardian contact is also a duplicate-aware shared number.
Preferred country / study level	Optional	Multi-select.
Preferred course, intake, university	Optional	Free text until Phase 2 structures them.
Highest education; education records; test scores	Optional	Structured rows (FR-24).
Work experience	Optional	Free text.
Previous visa refusal	Optional	YES/NO/UNKNOWN; highlighted on Person 360 when YES.
Lead source	Yes	Source → Campaign → Source Detail.
Branch; owner	Yes (active leads)	Single owner; history retained.
Stage; lead status; temperature	Yes	Per Section 10.
Lost reason; hold-until	If LOST / ON_HOLD	Controlled list; resume date.
Best contact method	Optional	WhatsApp or Call.
Consent	Yes for intake; recommended otherwise	Source, date, wording version, channels.
Phone verification	System	verified_at and method (OTP_SMS / OTP_WHATSAPP / STAFF_CALL / STAFF_IN_PERSON); unverified numbers are labeled in the UI.
Tags	Optional	Supplementary only; never replace structured fields.

Free-text 'notes' are stored as internal-note Activities rather than an editable field, so history is never lost.

23. Duplicate Detection

Normalization

- Store the raw phone and an E.164 value, produced with a phone-number library using default region NP; convert local Nepal formats to +977 where deterministic; support other country codes.
- Normalize emails to lowercase and trim whitespace; do not strip provider dots or plus-tags.
- Compare names case-insensitively after removing honorifics and extra spaces; transliteration and token order are similarity signals (trigram indexes).

Confidence model

Signal	Confidence	Behavior
Exact phone + exact email on the same record	EXACT	Very strong warning; always goes to review.
Exact normalized phone (not shared)	HIGH	Strong duplicate warning.
Exact email	HIGH	Strong duplicate warning.
Phone flagged shared (e.g. guardian)	MEDIUM	Require name/DOB/context review.
Name + DOB	HIGH	Strong possible-duplicate warning.
Name + another matching field	MEDIUM	Review warning.
Name only	LOW	Soft suggestion only.

- **Never auto-merge:** no confidence level, including EXACT, merges records automatically. EXACT and HIGH go to the review queue, MEDIUM is flagged on the record, LOW appears only as a suggestion at entry time and is not queued. Merging is always a human decision.
- Checks run on manual creation, QR/API intake, contact edits and import. Warnings never silently block: staff may proceed with a reason, and the override is audited.
- Unverified QR submissions are excluded from matching and never create, change or reveal anything about an existing Person; matching happens only after OTP verification.
- **Masked discovery (D-02):** when the match belongs to someone else, the counselor sees exists/owner/branch/status/date and a 'Request access' action instead of creating a second record. Confirming existence is a deliberate small disclosure; contact details stay hidden.
- High-confidence import and intake matches go to review rather than automatic merge. Only authorized users merge, after side-by-side comparison.
- Merge moves activities, follow-ups, tasks, documents, applications, payments and contact methods to the survivor; nothing is silently deleted. The merged record is archived with a pointer to the survivor, and Super Admin can reverse a merge within a defined window.

24. Follow-ups, Tasks, SLAs & Notifications

Concept	Definition and behavior
Follow-up vs Task	A FollowUp is a planned contact with the person (call, WhatsApp, email, meeting, walk-in, payment reminder). A Task is internal work (request a document from the registrar, prepare an application, check a deadline). Document requests and application updates are Tasks; contacting the person about them is a FollowUp.
Status	Open, in progress, completed, cancelled. Overdue is derived from due date/time.
Outcome	Controlled outcome plus notes; completing a follow-up prompts creation of the next one.
Reminder	Phase 1: computed in-app from due dates; Phase 5 adds email/WhatsApp/SMS.
No next action	Active leads with no open follow-up appear in a dedicated dashboard queue.
Notification	Persistent in-app alert: recipient, type, title, message, read state, related object. Triggers: assignment, intake received, overdue, duplicate review, access request, reassignment decision.
Unified queue (later)	Eventually combines follow-ups, tasks, application deadlines, offer expiry, visa appointments and document expiry.

Next action

- Next action is the earliest open FollowUp or Task flagged as next action for the Person. It is cached on Person (type, due time, owner) and kept in sync by the service layer so lists can sort and filter without joins; it is a projection, not a separate source of truth (D-25).
- Counselor views answer 'what do I do next for this student?'; leads without a next action are flagged.

Business calendar

ConsMan keeps a BusinessCalendar per branch: time zone Asia/Kathmandu, working days and hours, public and branch holidays. Default (to confirm): Sunday-Friday 10:00-17:00, Saturday closed. All business-hour due times are computed from it, and a calendar change recomputes open timers.

SLA rules

Rule	Trigger → target event	Default
First contact (new lead)	Person created → first logged contact	24 business hours
First contact (HOT lead)	Temperature set to HOT → first logged contact	4 business hours
QR lead assignment	Submission verified → owner assigned	2 business hours
Access request response	AccessRequest created → approved or denied	8 business hours

SLAs are configurable rules in Settings, not code constants. A timer starts on its trigger, runs only in business hours and is satisfied by its target event; on breach it raises a flag, notifies the owner and escalates to the Branch Manager on the dashboard.

25. Activities & Communication Logging

An Activity is a historical event, including internal notes. Activities are append-only: to fix a mistake a user adds a correction linked to the original (corrects_id); the timeline shows the corrected version with an 'edited' marker, and the original remains in history. Attachments are stored as Document records so permissions and audit apply once.

Field	Purpose
Type	Call, message, meeting, note, status change, assignment, intake, document event, payment event, system event.
Channel / Direction	WhatsApp, SMS, email, Messenger, phone, meeting, system / inbound, outbound, internal.

Field	Purpose
Outcome / Subject / Notes	Reached, no response, interested, requested callback, document received...; short context; detail.
Performed by / at	Staff or system actor and timestamp (UTC, shown in NPT).
Related	Person and optional Application, Task or Document.

Activity vs Communication vs Audit Event

Concept	What it is	Examples	Stored in
Activity	A business interaction or outcome recorded by staff or the workflow.	Called student; counseling completed; UK options discussed; internal note.	Activity
Communication	The actual message and its delivery state. Phase 1: manually logged messages are Activities with a channel, and OTP messages are MessageDelivery rows. Phase 5 adds a full Communication log.	WhatsApp sent; email received; SMS delivered.	MessageDelivery (Phase 1), Communication (Phase 5)
Audit Event	A system or security action: who changed what, when.	Owner changed; data exported; duplicates merged; phone edited; permission changed.	AuditEvent (append-only)

An action may legitimately produce both an Activity and an Audit Event (for example reassignment writes an assignment Activity for the timeline and an AuditEvent for accountability), but staff-facing history lives in Activity and accountability evidence lives in AuditEvent.

Unified Person timeline

Person 360 shows one chronological timeline built as a read model (not a stored table) from: Activities; FollowUp and Task events (created, completed, overdue); intake and verification milestones (registered via QR, OTP verified, Person created); assignment, status, stage and temperature changes; later, application, document, offer, payment and visa events. Security-only audit events such as exports or permission changes never appear. Each entry shows time (NPT), actor, type, summary and a link to the object; entries respect the viewer's permissions and can be filtered by type.

26. Dashboards & Reporting

Counselor

New leads assigned today · open leads by temperature · follow-ups due today · overdue follow-ups · active leads without next action · tasks due today · recent activity · applications and deadlines in later phases.

Branch Manager / Management

Intake queue (unassigned, aging beyond 2 hours) · QR funnel per code (scans → submissions → verified → Persons) and OTP verification rate · unverified queue size and age · lead volume by period · source/campaign/source-detail distribution (reportable independently) · pipeline with lost reasons · counselor workload · overdue follow-ups · time to first contact · lead → counseling → application → enrollment conversion · pending access/reassignment requests · later: applications by country/university/intake, offer/visa pipeline, finance summaries.

27. Audit & Accountability

- Audit foundation is built in Sprint 0 so every later feature writes audit events from day one.

- Record creation, updates, assignment/status changes, overrides, conversions, merges, exports, bulk actions, imports and permission changes: actor, timestamp, action, object, old/new values, request id.
- **Append-only:** the application database role has INSERT and SELECT only on the audit table; no UI exists to edit or delete audit rows.
- Exports and bulk actions record filters and record count. Financial and visa actions have stronger controls.
- Sensitive-document views, downloads, generated links and revocations are logged.
- QR create/edit/pause/archive, manual verification by call, discards, OTP lockouts and provider-limit events are audited.

28. Security & Privacy

- Role permissions plus object and branch scope, enforced server-side on every endpoint and queryset (Section 8).
- Authentication: Django session auth over HTTPS with secure, HttpOnly, SameSite cookies and CSRF protection; password policy and login rate limiting; MFA required for Super Admin and Finance.
- Sensitive documents are served by short-lived signed URLs, never predictable public paths; type and size limits and virus scanning on upload.
- Encrypt data in transit and at rest, including backups. Secrets live outside source control. Separate environments for development, staging and production; production data is not used in development.
- Records are ACTIVE or ARCHIVED. Normal users cannot hard-delete Persons or other business records; hard deletion happens only through a controlled retention or erasure process that needs Super Admin approval and writes an AuditEvent. Merge, archive, permission and workflow changes need elevated permission.
- Consent and retention: consent records on every intake path; configurable retention policy (D-08); anonymization/deletion procedure on valid request; per-channel opt-out before Phase 5 automation.
- Public intake: store-first durability, OTP controls and rate limits (Section 15), Turnstile, isolation of unverified data, and no personal data in logs, alerts or notifications about unverified submissions.
- Legal/privacy requirements must be reviewed against applicable Nepal law and destination-country obligations by counsel. This PRD defines design requirements, not legal advice.
- Define breach-response contacts and process before go-live.

29. Ownership, Assignment & Visibility

Situation	Rule
Counselor views leads	Full details for own assigned records; masked existence for others (D-02).
Existing-record conflict	Counselor raises an AccessRequest (access or reassignment) instead of creating a second record.
Branch Manager	Sees all records in own branch; cross-branch access only by grant.
Reassignment	Counselor requests; Branch Manager approves; managers may reassign directly. Records from/to, reason and timestamp in OwnershipHistory.
Bulk reassignment	Manager or Super Admin only; audit logged.
Departing staff	Deactivation is blocked while the user owns open records; bulk reassign first.
Export	Counselors export own scope; managers branch scope; sensitive fields excluded unless permitted.
Finance data	Finance Staff, Branch Manager, Management and Super Admin only, by default.

Assignment algorithm

- 1. The QR resolves to a Branch. 2. The submission is verified. 3. The branch's AssignmentRule is read.
- 4. **ROUND_ROBIN:** pick the next eligible counselor (active, available, in the branch, below the max-open-leads limit). **MANAGER_QUEUE:** place the Person in the branch's unassigned queue and notify managers.
- 5. Create the first follow-up. 6. Start the SLA timers (D-22, D-23). Every automatic assignment writes OwnershipHistory with the reason 'AUTO: round-robin'.

Unavailable owners

Situation	Behavior
Inactive or deactivated	Deactivation is blocked while the user owns open records; bulk reassign first.
On leave	Skipped by round-robin for the leave period; their open leads' overdue follow-ups surface to the Branch Manager, who may reassign (optional auto-reassign after a configurable number of days).
Transferred branch	Open leads stay with the owner only if a manager confirms; otherwise reassigned or the Person transfers branch, with audit.
Overloaded	Skipped once max_open_leads is reached; managers are alerted when every eligible counselor is at the limit.
No eligible counselor	Person goes to the manager queue with a notification; the assignment SLA keeps running.

30. Document Architecture

- **Person-level:** reusable identity/academic records (passport, citizenship, transcripts, IELTS, CV). **Application-level:** documents for a specific application, offer or enrollment process. **Visa-level:** financial evidence, TB certificates, forms, biometrics/interview evidence.
- A Document links to a Person and optionally an Application or VisaCase, with type, status, version, uploaded_by, verified_by, timestamps, expiry date and storage key.
- Versioning preserves replaced files; nothing is overwritten. Files live in object storage, never in PostgreSQL.
- Documents are not built in Phase 1 beyond the Document entity foundation and permission hooks (Section 38 rule).

31. Non-Functional Requirements

Area	Requirement
Performance	At Section 6 scale: search and list filters < 1 s (p95); dashboards and pages < 3 s (p95); import of 5,000 rows validated in < 2 min; public submit (store + trigger OTP) responds < 2 s (p95); OTP delivered in < 15 s median (provider dependent); QR generation and decode test < 5 s.
Availability	99.5% during business hours; maintenance outside business hours.
Recoverability	Automated daily database backups; RPO ≤ 24 h, RTO ≤ 4 h; restore tested before go-live and at least quarterly.
Durability	Once the server confirms a public submission, it survives provider outages, restarts and visitor abandonment (transaction committed before any provider call).
Security	As Section 28; OWASP ASVS L2 as the review baseline; dependency and secret scanning in CI.
Data integrity	Foreign keys, transactions, constraints and indexes on normalized contact values; service-layer validation.
Auditability	As Section 27.
Scalability	One branch to multiple internal branches without redesigning core entities.
File storage	S3-compatible object storage; size/type limits; virus scanning; signed URLs.
Observability	Structured logs, error tracking, health checks and uptime monitoring; request ids carried into audit events.
Accessibility and usability	Responsive UI; public form works on low-end phones and slow networks; keyboard accessible; WCAG 2.1 AA target for the public form.
Maintainability	Modular domain code, OpenAPI documentation, automated tests, reproducible deployments.

32. Technical Architecture

Layer	Decision
Frontend	Next.js + TypeScript + Tailwind; desktop-first staff UI; separate public intake route group with minimal bundle; typed API client generated from the OpenAPI schema.
Backend	Django + Django REST Framework, domain-oriented apps with services/selectors layers; PostgreSQL (with pg_trgm for fuzzy name search); drf-spectacular for OpenAPI; phone parsing with a maintained phone-number library.
API conventions	Versioned /api/v1; cursor or page pagination; consistent error body with code, message and field errors; request id on every response; public routes under /api/public. Idempotency-Key is supported on high-risk writes: public intake, OTP resend, payment creation, application submission, document upload, import commit and bulk operations.
Phase 1 async	No Redis/Celery. Overdue and stale flags are computed at query time; OTP challenges and rate limits use PostgreSQL tables with atomic counters; the provider call is made inline with a short timeout and a recorded failure path; nightly jobs (stale flags, unverified purge, intake payload purge, backups) run via host scheduler or management commands.
QR generation	Server-side in the qr domain: a maintained QR library plus a styling renderer (module and eye shapes, gradient, center logo, frames) producing PNG, SVG and PDF; automatic decode verification before download; assets versioned in object storage; form served at /r/{code} on the ConsMan domain, never through a third-party shortener.
Messaging provider	One MessagingProvider interface with SMS and WhatsApp adapters, delivery-status handling, secrets in environment variables and per-provider cost caps (D-19).
Phase 5 async	Introduce Redis + Celery for outbound messaging, webhooks and scheduled automation.
Storage and hosting	S3-compatible object storage (e.g. Cloudflare R2). Initial hosting (Cloudflare/Render/Neon or similar) subject to latency from Nepal and data-residency review.
Environments and delivery	Dev, staging, production; CI runs lint, type-check, unit and API tests, migrations check and security scans; infrastructure and config as code; seed data for roles, permission matrix and masters.
Testing	Permission-matrix tests generated from the matrix configuration; duplicate-detection test corpus of real Nepal phone formats; import idempotency tests; end-to-end tests for intake and merge.

Django domains

accounts · organizations (branches) · crm (person, contacts, consent, education/tests, activities, follow-ups, tasks, notifications, duplicates, access requests) · intake (submissions, verification) · qr · messaging · imports · catalog · applications · documents · offers · immigration · finance · communications · reports · audit

Each domain owns its models, services, permissions and API endpoints. Cross-domain access goes through services, not direct model imports of another domain's internals.

33. Key Screens & Information Architecture

Navigation: Dashboard · CRM (All People, My Leads, Intake Queue, Follow-ups, Tasks, Duplicate Review, Access Requests, Import Center) · Admissions · Visa & Enrollment · Finance · Communications · Reports · Settings.

Screen	Purpose	Phase
Public Visitor Form	Form → OTP entry → success; states for wrong/expired/locked code, send failure and inactive QR.	1
Login; Main Dashboard	Authentication; role-specific overview.	1
All People / My Leads	Search, filter, assign and manage.	1
Intake Queue	Tabs: Verified, Needs review, Unverified. Triage, assign, merge or create; verify by call or discard unverified.	1

Screen	Purpose	Phase
Person 360	Identity, owner, stage/status/temperature, next action, timeline, journey, applications, documents, payments where permitted.	1+
Follow-up Queue; Tasks	Today, upcoming, overdue; internal work.	1
Duplicate Review; Access Requests	Side-by-side compare and merge; approve or deny.	1
Import Center	Mapping, validation, duplicate review, import, revert.	1
Settings	Users, roles, branches, masters, retention, business calendar, SLAs, assignment rules, OTP parameters and messaging providers.	1
QR Studio (Settings → QR Codes)	Create, style, preview, test-scan, download and manage QR codes; per-QR analytics.	1
Admissions screens	Universities, Courses, Intakes, Applications, Application 360, Document Center, Offer Center.	2-3
Visa & Enrollment; Finance; Reports	Visa Center, Enrollment Documents, Pre-departure; Payments, commission; reporting.	3-4
Student Portal	Student-facing status and document actions.	6

LEAD → COUNSELING → CONVERTED (STUDENT) → APPLICATION → DOCUMENTS → OFFER → DEPOSIT → ENROLLMENT
DOCUMENT → VISA → PRE-DEPARTURE → ENROLLMENT

34. MVP Definition

The MVP is Phase 1: all requirements marked **M** in Section 12 (FR-01 to FR-43). Phase 1 delivers:

- Authentication, roles, branches and the permission matrix; audit and activity foundation.
- Person CRUD with multiple contact methods, source/campaign/tags, consent and temperature.
- Duplicate detection, merge, ownership, masked discovery and reassignment workflow.
- System-generated, customizable shared QR codes (QR Studio); OTP-verified visitor intake with store-first durability, intake queue and unverified queue.
- Follow-ups, tasks, notification center, activity timeline and manual communication logging.
- Search, filters and dashboards; Excel import, export and cutover.

Success criterion: staff manage active leads and follow-ups entirely inside ConsMan without parallel Excel trackers, and walk-in visitors register through the QR form instead of paper.

35. Acceptance Criteria — Phase 1

ID	Criterion	Traces to
AC-01	A new Person receives an immutable UUID and a BE-YYYY-NNNNNN reference and can be assigned a branch and owner; there is exactly one owner per active lead.	FR-01, 02, 04
AC-02	Phone formats 9801234567, 09801234567 and +977 980-123-4567 normalize to the same E.164 value and trigger a duplicate warning before save.	FR-05
AC-03	A shared-number match shows a lower-confidence warning and can be saved with a logged reason; name-only matches never block.	FR-05
AC-04	A counselor can create a follow-up with date/time, method and outcome in ≤ 3 clicks from Person 360.	FR-07
AC-05	Overdue follow-ups appear in the Overdue list and dashboard within 1 minute of becoming overdue; leads without a next action appear in their own queue.	FR-08
AC-06	Person 360 shows reverse-chronological history including manual communication entries; a corrected activity shows an 'edited' marker and keeps the original.	FR-09, 10
AC-07	Click-to-chat opens WhatsApp with the person's number and prompts to log an activity.	FR-11
AC-08	Merge moves 100% of activities, follow-ups, tasks, contacts and documents/applications to the survivor, archives the merged record with a pointer, and writes a MergeRecord and audit event.	FR-06
AC-09	A QR submission with a new phone, once its OTP is verified, creates a Person (LEAD/NEW) with ConsentRecord, source/branch from QR parameters, an intake Activity and a first-contact follow-up; it appears in the branch unassigned queue and notifies managers.	FR-12, 27, 20
AC-10	A verified QR submission matching an existing high-confidence record creates no second Person, adds an Activity, notifies the owner and creates a Duplicate Review item; the visitor sees the same confirmation as in AC-09.	FR-12, 05
AC-11	The public endpoint rejects requests without a valid Turnstile token, over rate limits, oversize payloads or filled honeypots, and returns the original result for a repeated Idempotency-Key.	FR-12
AC-12	An unassigned submission older than 2 business hours is flagged on the manager dashboard.	FR-12, 29
AC-13	Tasks can be created, assigned, prioritized and completed; completed tasks leave the open list.	FR-13
AC-14	Search by any phone format, email, name fragment or reference returns results in < 1 s (p95) at 20,000 records.	FR-14
AC-15	A manager can filter by country, counselor, branch, status, temperature, source/campaign, tag, date and follow-up state with the same performance target.	FR-15, 23, 28
AC-16	Excel import shows column mapping and valid/invalid/duplicate/needs-review counts before any write; re-importing the same file creates no duplicates; an unedited batch can be reverted as a unit.	FR-16
AC-17	Export is limited to the user's scope, excludes sensitive fields unless permitted, and creates an audit record with filters and count.	FR-17
AC-18	LOST without a reason and ON_HOLD without a resume date are rejected; ON_HOLD auto-creates a follow-up; backward moves require a reason.	FR-18
AC-19	Conversion changes stage LEAD $\rightarrow$ STUDENT on the same Person ID without creating a record, keeps all history and sets lead_status CONVERTED.	FR-18
AC-20	Bulk reassign/status/tag updates apply to selected records and create per-record audit entries.	FR-19
AC-21	Assignments, intake, overdue work, duplicate reviews and access requests generate in-app notifications with read state.	FR-20

ID	Criterion	Traces to
AC-22	A counselor opening another counselor's person sees only the masked view; requesting access creates an AccessRequest that a manager can approve or deny, recording OwnershipHistory.	FR-21, 04
AC-23	The staff-assisted form creates records through the same pipeline and additionally captures counseling summary, next step and temperature.	FR-22
AC-24	Education records and test scores can be added, edited and displayed on Person 360.	FR-24
AC-25	Automated tests generated from the permission matrix pass for every role/action pair; direct API calls cannot bypass UI restrictions.	FR-25
AC-26	Status change, reassignment, conversion, merge, export, import, bulk action and permission change each create an append-only audit record with actor, time and old/new values; the app DB role cannot update or delete audit rows.	FR-26
AC-27	Stale leads (no activity for the configured period) are flagged and listed; they are not closed automatically.	FR-30
AC-28	Backup restore to a clean environment is demonstrated before go-live.	NFR
AC-29	QR Studio creates a QR for a branch/source/campaign; PNG, SVG and PDF downloads are enabled only after the automatic decode test passes; scanning opens the form at register.theblessingedu.com/r/{code}.	FR-31, 32
AC-30	A center logo (default mark or upload) is placed at $\leq 20\%$ of QR area with error-correction level H applied automatically; designs that fail decoding are blocked with a reason.	FR-32
AC-31	At least 5 module styles, 4 eye-frame styles, 4 eye-ball styles, brand color presets, gradient, and caption/border frames are available; inverted or low-contrast ($< 4.5:1$) designs are blocked.	FR-32
AC-32	Registration, WhatsApp, URL and contact-card QR types each decode to the expected payload.	FR-33
AC-33	Restyling a QR keeps its code and URL unchanged; pausing or expiring it shows the friendly inactive page and accepts no submissions; the same QR can be scanned by many visitors producing independent submissions.	FR-34
AC-34	Submit commits an UNVERIFIED submission before the provider is called; with the provider disabled the submission still exists with reason SEND_FAILED and the visitor sees a 'details saved' message.	FR-36, 40
AC-35	A correct OTP within 5 minutes marks the submission and number verified (verified_at, verified_via) and the pipeline creates or matches the Person with a valid ConsentRecord.	FR-35
AC-36	Wrong codes: attempt count shown, locked after 5; expired codes rejected; resend blocked for 60 s and after 3 sends; OTPs exist only as hashes and never in logs.	FR-35, 38
AC-37	An unverified submission never creates, updates or matches a Person, never notifies an owner with personal data, and is visible only to Super Admin and the branch's Branch Manager.	FR-36, 37
AC-38	A manager can verify an unverified submission by call (reason required, audited) or discard it; unverified items are purged after 30 days.	FR-37
AC-39	Limits per phone, IP, QR and day are enforced; requests without a valid Turnstile token send no OTP; the spend cap raises an alert and throttles sending.	FR-38
AC-40	A visitor can correct the number up to twice, receives a new challenge each time, and can resume verification after a refresh with the resume token; original values remain in audit.	FR-35
AC-41	The QR list shows scans, submissions, verified submissions and Persons created per code, and the manager dashboard shows the verification rate.	FR-39
AC-42	A submission left unverified never creates a Person: the Person table is unchanged after UNVERIFIED submissions, and no temporary Person exists for them.	FR-35, 36

ID	Criterion	Traces to
AC-43	With a Sunday–Friday 10:00–17:00 calendar and a holiday, a 24-business-hour SLA started late on a working day is due at the correct time; changing the calendar recomputes open timers.	FR-42
AC-44	An SLA breach raises a flag, notifies the owner and escalates to the Branch Manager; timers do not run outside business hours.	FR-42
AC-45	A verified QR submission in a round-robin branch goes to the next eligible counselor (skipping on-leave, inactive and over-limit users); with none eligible it lands in the manager queue; OwnershipHistory records the automatic reason.	FR-43
AC-46	A Person's next action equals its earliest open follow-up/task and updates when that item is completed or rescheduled; leads without one are listed.	FR-41, 08
AC-47	The Person timeline shows activities, follow-up/task events, intake and OTP milestones and assignment/status changes in time order, hides security-only audit events and respects the viewer's permissions.	FR-09
AC-48	Import batches move STAGING → VALIDATED → APPROVED → COMMITTED; each imported Person shows its batch, row and source file; rolling back an unedited batch archives its records and writes audit.	FR-16
AC-49	Lost reasons come from the configurable master; LEAD → LOST is rejected without one; deactivating a reason keeps historical data intact.	FR-18
AC-50	No automatic merge occurs at any confidence level: EXACT/HIGH go to review, MEDIUM is flagged, LOW is only suggested at entry.	FR-05, 06

36. Data Migration & Import

- Collect all Excel sources into a controlled workspace; keep originals read-only as evidence.
- Define a canonical mapping and legacy-value tables: counselor names → users, free-text statuses → controlled statuses, source names → Source/Campaign masters, branch names → branches.
- Normalize phones, emails and names; keep raw values. Detect duplicates within files and against existing data; handle shared family numbers per Section 23.
- Generate a valid/invalid/duplicate/needs-review report before writing anything.
- **Idempotent:** each row carries source file, row number and optional legacy ID; re-running updates or skips rather than duplicating.
- **Batch lifecycle:** STAGING → VALIDATED → APPROVED → COMMITTED, and ROLLED_BACK when reverted. Every imported record carries import_batch_id, import_row_number and source_file, so 'where did this student's data come from?' is always answerable. A committed batch can be rolled back as a unit if its records have not been edited since; otherwise affected records are flagged for manual review.
- Import to staging first; dry-run on a copy; Branch Manager sign-off before the production load.
- Define the cutover date; after it, daily tracking happens only in ConsMan and spreadsheets become read-only. Paper forms are retired in favor of the QR form.

37. Risks & Remaining Confirmations

Risk	Mitigation
Staff keep using Excel or paper in parallel	Cutover date, read-only spreadsheets, QR codes at every desk, training, manager review of dashboards.
Poor legacy data quality	Early data audit, mapping tables, staging and human review.
Lead-ownership conflict	Masked discovery, access requests and clear approval flow; management sponsorship.
Public form abuse or spam	Turnstile, rate limits, honeypot, review queue; submissions never auto-merge.
OTP fraud (SMS pumping) and cost overruns	Turnstile before first send, per-phone/IP caps, country allow-list, resend limits, spend cap and alerts.
Provider outage or WhatsApp template delay	Two channels behind one interface, store-first so no data is lost, staff verify-by-call fallback, onboard providers before Sprint 3 ends.
Visitors cannot receive the OTP (no signal, typo, no WhatsApp)	Resend, switch channel, correct number, details saved, staff-assisted front-desk fallback.
Unverified data abused (someone enters another person's number)	Unverified data isolated from Persons, no matching or merging, short retention, staff verify-by-call.
Scope creep	MoSCoW and the Section 38 development rule.
Sensitive-data exposure	Least privilege, encryption, access logging, retention policy, legal review.
Hosting/connectivity from Nepal	Latency test in Sprint 0, backups, monitoring, low-bandwidth public form.

Remaining confirmations (do not block Sprints 0-2)

- Management sign-off on the Decision Log defaults (Section 7).
- Actual user, branch, monthly-lead and Excel-record counts to validate Section 6 and the performance targets.
- Legal review of retention values and consent wording (D-08, Section 28).
- Office WhatsApp/email addresses and the QR print set (branches, events).
- Select and contract the SMS and WhatsApp OTP providers; start the WhatsApp authentication-template approval early (needed before Sprint 4).
- Confirm OTP parameters, allowed country codes and the monthly OTP spend cap (D-17).
- Confirm the production domain for the public form (register.theblessingedu.com) and DNS/hosting control.
- Final permission matrix review by each role owner.
- Confirm office working days and hours, the public-holiday list and the SLA targets (defaults in D-23).

38. Recommended Development Order

Development rule: do not build later-phase features into Phase 1 merely because their models are foreseeable. Establish stable extension points but keep the MVP focused.

Step	Deliverable	Exit check
Sprint 0	Repo, CI, environments, Django/Next.js skeleton, auth, MFA, users, roles, branches, permission engine from the matrix config, audit foundation, Activity foundation , request ids, logging, backup and restore test, Nepal latency check, business calendar and SLA engine foundation .	AC-25, AC-26 skeleton, AC-28, AC-43
Sprint 1	Person, ContactMethod, source/campaign/tag masters, consent, education/tests, E.164 normalization, duplicate detection and masked view.	AC-01-03, 24

Step	Deliverable	Exit check
Sprint 2	Follow-ups, tasks, notifications, activity timeline, manual communication logging, click-to-chat, lifecycle rules and conversion, next action, SLA timers and escalation, unified timeline.	AC-04-07, 13, 18-19, 21, 44, 46, 47
Sprint 3	Search and filters, ownership/reassignment and access requests, merge, bulk actions, assignment engine , counselor and manager dashboards. Messaging providers onboarded (SMS + WhatsApp authentication template).	AC-08, 14-15, 20, 22, 27, 45, 49, 50
Sprint 4a	Public form at /r/{code}, store-first submission, OTP verification (SMS + WhatsApp), unverified queue, abuse and cost protection, intake queue, staff-assisted mode.	AC-09-12, 23, 34-40, 42
Sprint 4b	QR Studio: QR records, styling renderer, center logo, frames, types, decode test, PNG/SVG/PDF downloads, analytics.	AC-29-33, 41
Sprint 5	Excel import, export, migration dry-run, UAT with real data, cutover.	AC-16-17, 48
Sprint 6	Hardening, performance tuning to targets, security review, training, go-live.	All AC
Later	Phase 2 masters and applications → documents → Phase 3 offers, basic payments, enrollment documents, visa → Phase 4 finance → Phase 5 integrations and Redis/Celery → Phase 6 portal → Phase 7 AI.	Per-phase PRD addendum

39. Future Enhancements

- Advanced analytics and cohort reporting; calendar integration; document OCR and extraction.
- Mobile/PWA experience; Nepali-language UI; destination-specific workflow designer; saved views.
- AI-assisted counseling and document workflows; richer communication-provider synchronization.
- Student journey health indicator (On Track / Attention Required / Blocked) derived from objective rules over blockers, deadlines, missing documents and expired offers; marketing ROI attribution beyond QR funnels; AI prioritization.

40. Product Principles

- **Single source of truth:** no parallel spreadsheets or paper for normal operations.
- **Person-centric:** the Person is the central entity; applications are related records.
- **History matters:** never lose previous interaction context.
- **Workflow over memory:** deadlines and next actions are system-managed.
- **Application independence:** each application has its own state and history.
- **Configurable by destination:** UK, Australia, USA, Canada, South Korea, New Zealand, Cyprus and Europe can differ.
- **Human control:** automation assists staff; staff own consequential decisions.
- **Security by default:** sensitive data is protected and auditable.
- **Build incrementally:** every phase delivers usable value before the next.

41. Final Architecture

PERSON: VISITOR/LEAD → COUNSELING → CONVERTED (STUDENT) → APPLICATION(S) → DOCUMENTS → OFFER → DEPOSIT
→
CAS/CoE/I-20/LOA → VISA → PRE-DEPARTURE → ENROLLMENT

One continuous Person-centered record. Finance, communications, tasks, activities, notifications, reporting and audit operate across the journey. Applications remain independent: one Person may hold

several simultaneous applications with different universities, courses, intakes and states.

42. Glossary

Term	Meaning
Person	Single master record for anyone ConsMan tracks, from first contact through enrollment.
Lead / Student	A Person in stage LEAD or STUDENT; conversion is a controlled action on the same record.
Temperature	HOT/WARM/COLD priority, independent of lifecycle status.
IntakeSubmission	A stored public registration with its own verification status. Separate from a Person; a Person is created or matched only after verification.
QR code (record)	A system-managed QR with a permanent short code, branch/source/campaign, style and status; the same QR is shown to all visitors.
Unverified submission	A stored public submission whose phone has not yet been confirmed by OTP; isolated from Persons.
OTP challenge	A single-use, time-limited verification code attempt for one submission and number.
Business calendar / SLA rule	Working days, hours and holidays per branch, and configurable time targets computed from them.
Next action	Cached projection of the earliest open follow-up/task for a Person.
WorkflowTemplate	Versioned per-destination configuration of milestones, required documents and requirements.
Blocker / Deadline	A recorded obstacle on a Person, application or visa case / a dated obligation tracked as data.
Masked discovery	Limited view showing a record exists and who owns it, without contact details.
Application	A person's application to a university/course/campus/intake.
CourseOffering	A course instance tied to a campus and intake with requirements and commercial data.
Offer	Admission decision/offer associated with an application.
Enrollment Document	CAS, CoE, I-20, LOA or similar destination-specific document.
Visa Case	Structured immigration workflow for an application; re-applications are new attempts.
Activity / Follow-up / Task	Historical event / planned contact with the person / internal work item.
Institution Commission	Commission received by The Blessing Edu from a university; not agency/sub-agent commission.
E.164	International phone format, e.g. +9779801234567.
MoSCoW	Must / Should / Could / Won't prioritization.

End of PRD v1.5